Praveen Nair

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Education

2025 – **Brown University** – Providence, RI
Ph.D., Computer Science.
Advisor: Suresh Venkatasubramanian.
Currently: working on postdeployment monitoring of algorithmic decisionmaking systems.

2022 – 2024 **University of California, San Diego** – La Jolla, CA
M.S., Computer Science and Engineering. *GPA: 3.83.*
Thesis: "Active Learning and Epistemic Defenses of Fairness." Advisor: David Danks.

2018 – 2022 **University of California, San Diego** – La Jolla, CA
B.S., Data Science. *GPA: 3.961, Magna Cum Laude.*
Minors in history & linguistics, concentration in political science

Research Experience

Jan 2025 - Jun 2025 **Research Associate, Harvard Business School**
Advisor: Amit Goldenberg.
Worked on multiple studies and lab administration tasks. Implemented study procedure involving realtime human-AI and human-human interaction, conducted analysis, and contributed writing for project on human & AI performance in emotional support; in submission, on PsyArXiv.

Sep 2023 - Present **ION-C: Integration of Overlapping Networks with Constraints**
Continuing work with David Danks, Sergey Plis.
Answer-set programming algorithm for timeseries & stationary causal discovery with overlapping variables. Designed batched simulations, implemented causal learning algorithms, analyzed results. Problem formulated in ASP system *clingo*, work in Python and Slurm. Currently on arXiv.

March 2023 - Jun 2024 **Thesis: Active Learning and Epistemic Defenses of Fairness**
Advisor: David Danks.
Using active learning and causal modeling, showed that benign group differences can lead to uncertainty disparities in selective-labeling settings, with implications for high-stakes decision-making. Used Bayesian causal modeling, derived setting-specific formula for infomax active learning.

June 2021 - June 2022 **Stowers Lab, Scripps Research Institute**. PI: Lisa Stowers.
Primarily, used tools such as DeepLabCut and B-SOID on remote computing cluster to develop computer vision pipeline for behavioral analysis of video. Supported multiple projects on neurological underpinnings of physiological arousal, scent marking, olfaction, and behavior.

June 2021 - December 2021 **UC San Diego Computer Science & Engineering**. PI: Christine Alvarado.
Studied effects of early undergraduate CS research program on students' senses of identity as researchers and computer scientists. Used thematic analysis methods on open-ended reflection data, as well as Python for preprocessing, analysis, and interrater reliability calculation.

October 2020 - June 2021 **Qualcomm Institute @ UC San Diego**. PI: Leanne Chukoskie.
Survey project about student experiences with engagement in online learning at UCSD during COVID-19 adjustments. Co-first-author of paper published in Frontiers in Education, designed survey questions, analyzed survey and course review dataset in R, wrote paper Results section

Publications

2025 **Exploring Human and AI Emotional Support Through Reframing of Negative Situations**
Joanna Z. Li, Alina Herderich, **Praveen Nair**, Amit Goldenberg.
PsyArXiv preprint. https://osf.io/preprints/psyarxiv/fzvd8_v4

2024 **ION-C: Integration of Overlapping Networks via Constraints**
Praveen Nair, Payal Bhandari, Mohammadsajad Abavisani, Sergey Plis, David Danks.
arXiv preprint. https://arxiv.org/abs/2411.04243v1

2024 **Active Learning and Epistemic Defenses of Fairness.** (Master's thesis, UC San Diego.)
Praveen Nair. https://escholarship.org/uc/item/2hm001k0

2022 **Engagement in online learning: student attitudes and behavior during COVID-19**
Brooke Hollister*, **Praveen Nair***, Sloan Hill-Lindsay, Leanne Chukoskie. ***(*co-first-author)***
Frontiers in Education. https://doi.org/10.3389/feduc.2022.851019

Teaching Experience

Fall/Winter 2022-23 & 2023-24 **Teaching assistant, DSC 180A/B: Data Science Project.** Professor: Suraj Rampure.
Supervised undergraduate senior capstone projects, met with project groups, graded assignments, advised on course content, coordinated with academic and industry mentors.

Spring 2023, Spring 2024 **Teaching assistant, DSC 80: Practice and Application of Data Science.** Professors: Tauhidur Rahman & Sam Lau. Led discussion sections of 80-100 students with live coding, held very busy office hours, graded assignments, and wrote and graded portions of exams.

Other Experience

June 2020 - June 2022 **Sports Editor, UCSD Guardian**
Wrote over 75 articles for UCSD's campus newspaper, edited many more, created 71-minute documentary about campus history awarded by San Diego Press Club, featured by SD Union-Tribune.

Summer 2020 **Percolata, Software Engineering and Machine Learning Intern**
Software and machine learning experimentation for algorithmic trading product in Google Cloud Platform. Worked with Python, GCP, multiple timeseries prediction & deep learning frameworks.

Technical Skills

Languages: Primarily Python and R, also SQL (MariaDB), Java, Go, MATLAB, HTML/CSS, JS.
Software: LaTeX, Git, Docker/Kubernetes, AWS, Google Cloud Platform, Slurm, LSF, Clingo.
Research Methods: Simulation design, Qualtrics & Smartriqs, survey design & analysis, thematic coding, visualization